Biology Exam Language Bank - Say It Like the Mark Scheme (9BI0)

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Purpose: Jody understands the biology, but marks are awarded for specific scientific terms and cause-and-effect links, not for the general idea. This bank gives her the exact wording the mark scheme looks for, topic by topic, so she can learn to answer in that language.

用法簡述： 每個主題都用三欄對照：概念、「自己的話」答法（失分）、「考試用語」答法（得分）。先蓋住右欄自己寫，再對照、劃出漏掉的術語。

How marks are awarded

- **One mark per precise point.** A mark scheme is a list of separate points, and each earns one mark when the key term or idea appears. A long answer that makes no precise point can score zero.
- **Key terms matter.** "Denatured", "complementary", "hydrolysis" and "water potential" are usually the words the examiner is looking for.
- **Link the cause to the effect.** "Hydrogen bonds break, so the tertiary structure changes, so the active site is no longer complementary" earns more than "the enzyme gets damaged".
- **Vague words earn nothing.** "It", "stuff", "faster" and "damaged" are replaced by named structures and mechanisms (see the next section).
- **Quote data when asked to describe.** A trend without figures usually loses a mark.

How to use this bank

1. Read one row. Cover the right-hand column and write your own answer to the concept.
2. Uncover the exam-language answer. Underline every key term you missed.
3. Close the page and rewrite the answer using those terms. Repeat on a later day until the terms come without prompting.

The "own words" column shows typical answers that lose marks. After the mock paper is marked, replace them with Jody's own wording, which makes the contrast personal.

Command words: what each one asks for

Command word	What to write	Mark logic
State, Give, Name	One short fact or term, no explanation	1 mark for the correct term
Define	The accepted definition, with the key words	Each key word or idea is a mark point
Describe	What happens, in order, with no reasons. For data: the trend, quoted values and any change in direction	A mark for each stage or each quoted value
Explain	Why it happens: a point, then the biological reason, then the consequence	Marks for the reason and the link, not the description
Suggest	Apply known biology to an unfamiliar situation; there are several acceptable answers	Marks for a sensible idea that uses correct biology
Compare	Similarities and differences, each stated for both items	Marks only when both sides are given
Calculate	Formula, substitution, answer with units	Method mark and answer mark
Deduce, Predict	A conclusion drawn from the information given, with the reasoning if asked	Marks for a conclusion that follows from the data
Evaluate, Discuss	Advantages and disadvantages, then a reasoned judgement	Marks for balance and a justified conclusion
Justify	A claim with the evidence that supports it	A mark for the claim and one for the evidence

Vague words to replace

These everyday words are the commonest reason a correct idea scores nothing. Swap each one for the scientific wording.

Vague word or phrase	Write this instead
"it" (unclear which thing)	Name the structure or molecule every time: "the enzyme", "the active site", "the substrate"
"more", "less", "bigger", "faster"	"increases the rate of ...", "a greater proportion of ...", and quote the values with units
"damaged", "killed", "broken", "dies"	An enzyme is "denatured"; a polymer is "hydrolysed"; a cell "lyses"; a membrane "loses its partial permeability"
"stuff", "things", "substances"	Name them: "oxygen", "glucose", "mineral ions", "solutes"
"goes through", "passes across", "gets in"	"diffuses across ... down a concentration gradient", "moves by osmosis", "is actively transported"
"helps", "lets", "allows"	State the mechanism: "provides a large surface area", "shortens the diffusion distance"
"the heat"	"kinetic energy" (molecules) or "thermal energy"
"the enzyme fits the substrate"	"the substrate is complementary to the active site and forms an enzyme-substrate complex"
"more collisions"	"more frequent collisions between enzyme and substrate, forming more enzyme-substrate complexes per unit time"
"energy is made"	"energy is released or transferred"; "ATP is synthesised". Energy cannot be made
"cells breathe"	"cells respire"; "organisms ventilate their lungs"
"water is sucked up"	"water moves by osmosis down a water potential gradient"; "water is pulled up the xylem by transpiration pull"
"pushes"	"exerts pressure"; "hydrostatic pressure forces fluid out"
"concentration" used for the amount	Use "concentration" only for a ratio such as mol dm-3; otherwise say "amount" or "number"
"the blood carries oxygen"	"oxygen is transported bound to haemoglobin as oxyhaemoglobin"

Sentence frames to reuse

Frame	Pattern	Example
Structure to function	[Feature] gives [adaptation], so [function] happens by [mechanism].	Alveoli walls are one cell thick, which shortens the diffusion distance, so gases diffuse quickly.
Cause chain	Because [A], [B], so [C].	Because hydrogen bonds break, the tertiary structure changes, so the active site is no longer complementary.
Rate link	... so there are more frequent collisions, forming more enzyme-substrate complexes per unit time, so the rate of reaction increases.	Used for temperature, concentration and inhibitor questions.
Describing data	As [variable] increases from [a] to [b] [unit], [response] increases from [c] to [d] [unit], then [plateaus or falls].	As pH rises from 3 to 7, oxygen volume rises from 2 to 30 cm3, then falls to 6 cm3 at pH 9.
Comparing	Both [shared feature]; whereas [A has x], [B has y].	Both have double membranes; whereas mitochondria have cristae, chloroplasts have thylakoids.
Gradient	[Substance] moves from a [higher] to a [lower] [concentration, water potential, pressure] by [process].	Water moves from a higher to a lower water potential by osmosis.

Topic 1: Biological Molecules

Concept	Own words (loses marks)	Exam language (scores)
Condensation and glycosidic bond	Two sugars stick together and water comes out.	A condensation reaction joins two monosaccharides with a glycosidic bond and releases a molecule of water.

Concept	Own words (loses marks)	Exam language (scores)
Hydrolysis	Water breaks the bond.	Hydrolysis breaks a glycosidic, peptide or ester bond by adding a molecule of water.
Water as a solvent	Water is good at dissolving things.	Water is a polar molecule, so polar and ionic substances dissolve in it; this allows metabolic reactions to occur in solution and substances to be transported.
High specific heat capacity	Water doesn't get hot quickly.	Hydrogen bonds between water molecules absorb a lot of energy before the temperature rises, so water has a high specific heat capacity and temperature changes are buffered.
Cohesion and surface tension	Water sticks together.	Hydrogen bonds between water molecules cause cohesion, so water forms a continuous column in the xylem and has a high surface tension.
Starch	Starch is where plants store energy.	Starch is a storage polysaccharide of α-glucose made of amylose (unbranched, coiled and compact) and amylopectin (branched, with many ends for rapid hydrolysis). It is insoluble, so it does not affect water potential.
Glycogen	Animals store glycogen.	Glycogen is a highly branched polymer of α-glucose. The many terminal ends allow rapid hydrolysis to release glucose, and it is compact and insoluble.
Cellulose	Cellulose is strong.	Cellulose is a polymer of β-glucose in straight chains. Hydrogen bonds between parallel chains form microfibrils with a high tensile strength, which supports the cell wall.

Concept	Own words (loses marks)	Exam language (scores)
Triglyceride	A fat has three tails.	A triglyceride is one glycerol joined to three fatty acids by ester bonds, formed in condensation reactions.
Saturated and unsaturated fatty acids	Unsaturated fats are bendy.	Saturated fatty acids have no C=C double bonds; unsaturated fatty acids have one or more C=C double bonds, which cause kinks in the hydrocarbon chain.
Phospholipid	The head loves water and the tail hates it.	The phosphate head is hydrophilic and the fatty acid tails are hydrophobic, so in water phospholipids form a bilayer with the heads facing the water and the tails facing inwards.
Peptide bond	Amino acids join together.	A condensation reaction between the amine group of one amino acid and the carboxyl group of another forms a peptide bond and releases water.
Protein structure	The protein folds into a shape.	Primary: the sequence of amino acids joined by peptide bonds. Secondary: hydrogen bonds form α-helices and β-pleated sheets. Tertiary: further folding held by hydrogen bonds, ionic bonds, disulfide bonds and hydrophobic interactions. Quaternary: two or more polypeptide chains.
Structure and function of proteins	The shape is important.	The specific tertiary structure, determined by the primary sequence, gives a protein its function, for example the shape of an enzyme's active site.

Concept	Own words (loses marks)	Exam language (scores)
Enzyme action	The enzyme fits the substrate like a lock and key.	The substrate binds to the active site, which is complementary to its shape, forming an enzyme-substrate complex. Induced fit moulds the active site around the substrate, which lowers the activation energy.
Activation energy	The enzyme makes the reaction faster.	The enzyme lowers the activation energy by holding the substrates in the correct orientation and straining their bonds, so the reaction needs less energy to start.
Effect of temperature	Too hot and the enzyme dies.	As temperature rises, kinetic energy increases, so collisions are more frequent and more enzyme-substrate complexes form. Above the optimum, hydrogen and ionic bonds break, the tertiary structure changes, the active site is no longer complementary and the enzyme is denatured.
Effect of pH	The wrong pH damages the enzyme.	A pH away from the optimum changes the charges on the R groups and breaks ionic and hydrogen bonds. The tertiary structure and the shape of the active site change, so fewer enzyme-substrate complexes form.
Effect of substrate concentration	More substrate makes it faster until it stops.	Rate increases because more substrate gives more frequent collisions and more complexes. It levels off when all the active sites are occupied, so enzyme concentration becomes the limiting factor.

Concept	Own words (loses marks)	Exam language (scores)
Competitive inhibitor	It blocks the enzyme.	The inhibitor has a shape similar to the substrate, binds to the active site and prevents the substrate binding. The effect is reduced by increasing the substrate concentration.
Non-competitive inhibitor	It changes the enzyme.	The inhibitor binds to an allosteric site, which alters the tertiary structure and the shape of the active site so the substrate cannot bind. Increasing the substrate concentration does not overcome it.
DNA structure	DNA is a twisted ladder.	DNA is a double helix of two antiparallel polynucleotide strands. Nucleotides are joined by phosphodiester bonds, and complementary bases pair by hydrogen bonds: A with T by two, C with G by three.
DNA and mRNA	RNA is the copy.	DNA is double-stranded with deoxyribose and thymine. mRNA is single-stranded, shorter, and has ribose and uracil.
Semi-conservative replication	DNA copies itself.	Helicase breaks the hydrogen bonds and unwinds the helix. Each strand acts as a template. Free nucleotides pair by complementary base pairing. DNA polymerase joins them with phosphodiester bonds. Each new molecule has one original and one new strand.

Concept	Own words (loses marks)	Exam language (scores)
Meselson and Stahl	They proved DNA copies itself.	Bacteria grown in 15N were transferred to 14N. After one generation all the DNA had an intermediate density; after two, half was intermediate and half light. This supports semi-conservative replication.
Biochemical tests	Add the chemical and it changes colour.	Reducing sugar: heat with Benedict's reagent, blue to green, yellow, orange or brick-red precipitate. Starch: iodine solution, yellow-brown to blue-black. Protein: biuret reagent, blue to purple. Lipid: dissolve in ethanol then add water, cloudy white emulsion.
Test for a non-reducing sugar	Boil it again.	Boil with dilute hydrochloric acid to hydrolyse the glycosidic bond, neutralise with sodium hydrogencarbonate, then heat with Benedict's reagent. A brick-red precipitate shows a non-reducing sugar was present.

Key terms to include: condensation, hydrolysis, glycosidic bond, ester bond, peptide bond, hydrophilic, hydrophobic, complementary, active site, enzyme-substrate complex, activation energy, denatured, tertiary structure, allosteric site, antiparallel, phosphodiester bond, complementary base pairing, template.

Topic 2: Cells, Viruses and Reproduction

Membranes and transport are included here. Check your class notes for whether they were taught under Topic 1 or Topic 2; the wording is the same either way.

Concept	Own words (loses marks)	Exam language (scores)
Fluid mosaic model	The membrane has phospholipids and proteins.	The membrane is a phospholipid bilayer that is fluid because the molecules move laterally, with proteins embedded in it (the mosaic). Cholesterol regulates fluidity and glycoproteins act as receptors.
Simple diffusion	Things move from lots to little.	The net movement of molecules from a region of higher concentration to a region of lower concentration, down a concentration gradient, due to random movement. No energy is needed.
Facilitated diffusion	Proteins help things across.	Polar molecules and ions diffuse down their concentration gradient through channel or carrier proteins. No ATP is used.
Osmosis	Water goes to where there is more salt.	The net movement of water from a region of higher water potential to a region of lower water potential across a partially permeable membrane.
Active transport	It uses energy to move things.	The movement of substances against a concentration gradient through carrier proteins, using energy from ATP.
Endocytosis and exocytosis	The cell swallows things or spits them out.	In endocytosis the membrane invaginates and forms a vesicle around the material; in exocytosis vesicles fuse with the cell-surface membrane and release their contents.

Concept	Own words (loses marks)	Exam language (scores)
Prokaryote and eukaryote	Bacteria don't have a nucleus.	Prokaryotes have no membrane-bound organelles, circular DNA free in the cytoplasm, 70S ribosomes and a peptidoglycan cell wall. Eukaryotes have a nucleus, membrane-bound organelles and 80S ribosomes.
Nucleus	The nucleus is the control centre.	The nucleus contains DNA as chromosomes, which codes for proteins. Nuclear pores let mRNA leave, and the nucleolus makes ribosomal RNA.
Mitochondria	The powerhouse of the cell.	The site of aerobic respiration. The inner membrane is folded into cristae to give a large surface area for ATP synthesis, and the matrix contains enzymes.
Chloroplast	Where photosynthesis happens.	The site of photosynthesis. Thylakoid membranes stacked in grana contain chlorophyll to absorb light, and the stroma contains the enzymes for the light-independent reactions.
Protein secretion pathway	The cell makes the protein and sends it out.	Protein is made on ribosomes on the rough endoplasmic reticulum, transported in vesicles to the Golgi apparatus, which modifies and packages it into secretory vesicles. The vesicles fuse with the cell-surface membrane and release the protein by exocytosis.
Lysosome	It gets rid of waste.	Lysosomes contain hydrolytic enzymes that digest worn-out organelles and material taken in by endocytosis.

Concept	Own words (loses marks)	Exam language (scores)
Resolution and magnification	The electron microscope zooms in more.	Magnification is how many times larger the image is than the object. Resolution is the ability to distinguish two points that are close together. An electron microscope has higher resolution because electrons have a shorter wavelength than light.
Magnification calculation	Measure the picture and work it out.	Magnification = image size divided by actual size. Convert to the same units first: 1 mm = 1000 µm. Rearrange to find the actual size = image size divided by magnification.
TEM, SEM and light microscope	TEM is stronger than SEM.	TEM passes electrons through a very thin specimen and gives a high-resolution 2D image of internal structure. SEM scans electrons over the surface and gives a 3D image of the surface. Both need a vacuum, so the specimen must be dead. A light microscope can view living specimens but has lower resolution.
Staining	It makes it easier to see.	Stains bind to specific structures and increase the contrast, so different structures can be distinguished.
Cell cycle	The cell grows and then divides.	Interphase: G1 (cell growth, protein synthesis), S (DNA replication) and G2 (growth, preparation for mitosis). Then mitosis, followed by cytokinesis.

Concept	Own words (loses marks)	Exam language (scores)
Mitosis stages	The chromosomes line up and get pulled apart.	Prophase: chromosomes condense and nuclear envelope breaks down. Metaphase: chromosomes line up on the equator attached to spindle fibres. Anaphase: centromeres divide and sister chromatids are pulled to opposite poles. Telophase: nuclear envelopes reform and chromosomes decondense.
Mitotic index	The number of cells dividing.	Mitotic index = number of cells in mitosis divided by total number of cells.
Importance of mitosis	The cell makes copies of itself.	Mitosis produces genetically identical daughter cells with the same chromosome number, for growth, tissue repair and asexual reproduction.
Cancer	Cells grow out of control.	Mutations in the genes that control the cell cycle cause uncontrolled cell division, which forms a tumour.
Virus structure	A virus is a tiny germ.	A virus is acellular: genetic material (DNA or RNA) inside a protein capsid, with attachment proteins on the surface; some have a lipid envelope. It has no cytoplasm or organelles and can only replicate inside a host cell.

Concept	Own words (loses marks)	Exam language (scores)
HIV replication	HIV takes over the cell.	The attachment protein binds to the CD4 receptor on a T helper cell. The RNA enters the cell, and reverse transcriptase makes DNA from the RNA. The DNA is inserted into the host genome, host ribosomes make viral proteins, and new viruses are assembled and released by budding.
Fertilisation in mammals	The sperm meets the egg.	The acrosome releases hydrolytic enzymes that digest the egg's outer layer. The sperm and egg membranes fuse, and the nuclei fuse to form a diploid zygote.
Fertilisation in flowering plants	The pollen reaches the ovule.	A pollen grain lands on the stigma and a pollen tube grows down the style to the ovule. In double fertilisation, one male nucleus fuses with the egg nucleus to form the zygote and the other fuses with the polar nuclei to form the endosperm.

Key terms to include: phospholipid bilayer, partially permeable, water potential, carrier protein, channel protein, vesicle, cristae, resolution, wavelength, sister chromatids, spindle fibres, genetically identical, acellular, capsid, reverse transcriptase, zygote, diploid.

Topic 3: Classification and Biodiversity

Concept	Own words (loses marks)	Exam language (scores)
Taxonomic hierarchy	Animals are put into groups.	Domain, kingdom, phylum, class, order, family, genus, species: a hierarchy in which each group is nested inside a larger group and there is no overlap.

Concept	Own words (loses marks)	Exam language (scores)
Binomial naming	Scientists give animals a Latin name.	The binomial system names an organism by its genus then its species. The genus has a capital letter, the species a lower-case letter, and both are italicised or underlined. It is universal, so it avoids confusion caused by different common names in different languages.
Species	Animals that are the same.	A group of organisms with similar features that can interbreed to produce fertile offspring.
Three domains	Bacteria are split into two types.	The three domains are Archaea, Bacteria and Eukarya. They were proposed from differences in ribosomal RNA sequences and cell membrane chemistry, which showed that Archaea differ from Bacteria.
DNA evidence for classification	Scientists use DNA to see who is related.	Comparing DNA base sequences (or amino acid sequences of proteins) shows the more similar the sequences, the more recent the common ancestor. It is objective evidence that can resolve classifications based on appearance alone.
Phylogenetic tree	A family tree of animals.	A phylogenetic tree shows evolutionary relationships. Each branch point represents a common ancestor, and species that share a more recent branch point are more closely related.
Biodiversity	The number of different animals.	The variety of living organisms in an area. It includes species richness (the number of species), genetic diversity (the variety of alleles within a species) and ecosystem diversity.

Concept	Own words (loses marks)	Exam language (scores)
Index of diversity	A big number means lots of animals.	D = 1 minus the sum of (n/N) squared. A value close to 1 shows many species with similar numbers. A low value shows a few species dominate, and the community is more vulnerable to change.
Genetic diversity and survival	More variation makes them stronger.	A greater variety of alleles means that if the environment changes or a disease arises, some individuals are more likely to have an allele that gives resistance. They survive and reproduce, so the species is less likely to become extinct.
Causes of reduced genetic diversity	Small populations are weak.	A population bottleneck or the founder effect leaves fewer alleles in the gene pool, and inbreeding increases the proportion of homozygous recessive genotypes.
Random sampling	Pick places at random.	Use a grid or tape measures to divide the area, and use a random number generator to pick coordinates. This removes bias, so the sample is representative. Take enough samples and repeat.
Natural selection	The strongest survive.	Variation exists in the population because of mutation. A selection pressure, such as disease, acts on the population. Individuals with the advantageous allele are more likely to survive and reproduce, and they pass on that allele. Over generations the allele frequency increases.

Concept	Own words (loses marks)	Exam language (scores)
Antibiotic resistance	Bacteria learn to resist antibiotics.	A random mutation gives some bacteria resistance. The antibiotic is a selection pressure, so resistant bacteria survive and reproduce, passing on the resistance allele by vertical gene transmission. The allele frequency increases in the population.
Types of adaptation	The animal is built for its environment.	Anatomical (a structural feature, for example thick fur), physiological (a feature of how the body works, for example producing antifreeze proteins) or behavioural (an action, for example migration).
Evidence for evolution	There are fossils.	The fossil record shows changes in organisms over time. DNA sequence comparisons and protein comparisons such as cytochrome c show similarities that match the proposed evolutionary relationships.

Key terms to include: hierarchy, binomial, genus, species, fertile offspring, domain, ribosomal RNA, common ancestor, species richness, genetic diversity, allele, gene pool, bottleneck, founder effect, selection pressure, allele frequency, random sampling.

Topic 4: Exchange and Transport

Concept	Own words (loses marks)	Exam language (scores)
Surface area to volume ratio	Small animals have more surface.	Small organisms have a large surface area to volume ratio, so the diffusion distance is short and oxygen reaches all cells by diffusion. Large organisms have a small ratio, so they need specialised exchange surfaces and a mass transport system.

Concept	Own words (loses marks)	Exam language (scores)
Features of a good exchange surface	It has to be big and thin.	A large surface area, a thin barrier giving a short diffusion distance, and a steep concentration gradient maintained by ventilation and blood flow. These increase the rate of diffusion (Fick's law).
Insect gas exchange	Insects have holes in their body.	Air enters through spiracles into tracheae, which branch into tracheoles that reach the muscle cells. Oxygen diffuses down its concentration gradient directly to the cells, and abdominal pumping ventilates the system.
Fish gills	Water goes one way and blood goes the other.	Water and blood flow in opposite directions (countercurrent), so a concentration gradient is maintained along the whole length of the lamella. Oxygen keeps diffusing into the blood, and equilibrium is never reached, so a high percentage of oxygen is absorbed.
Alveoli adaptations	The lungs are good for gas exchange.	The alveolar wall and capillary wall are each one cell thick, giving a short diffusion distance. There are millions of alveoli, which give a large surface area. Ventilation and blood flow maintain a steep concentration gradient.
Inspiration	You breathe in and the lungs fill.	The external intercostal muscles contract, raising the ribs up and out, and the diaphragm contracts and flattens. The volume of the thorax increases, so the pressure falls below atmospheric pressure and air flows in.

Concept	Own words (loses marks)	Exam language (scores)
Expiration	You breathe out.	The external intercostal muscles relax and the ribs move down and in; the diaphragm relaxes and rises. The thorax volume decreases and the pressure rises above atmospheric, so air is forced out. Forced expiration uses the internal intercostal muscles.
Gas exchange in a leaf	Air gets in through the holes.	Gases diffuse through open stomata into the air spaces of the spongy mesophyll, then dissolve in the film of water on the cell walls and diffuse into the cells. Guard cells control the size of the stomata.
Xerophyte adaptations	Cacti keep water.	A thick waxy cuticle reduces evaporation. Sunken stomata, rolled leaves and hairs trap a region of humid air, which reduces the water potential gradient, so less water vapour diffuses out and the rate of transpiration falls.
Water loss and carbon dioxide uptake	The plant has to open its stomata.	Stomata must open for CO2 to diffuse in for photosynthesis, but opening them also lets water vapour diffuse out. The plant balances these two needs.
Artery, vein and capillary	Arteries are thick and veins have valves.	Arteries have thick walls with elastic fibres to withstand high pressure and recoil to maintain the pressure. Veins have thinner walls and valves to prevent backflow. Capillaries are one cell thick, giving a short diffusion distance.

Concept	Own words (loses marks)	Exam language (scores)
Tissue fluid formation	Fluid leaks out of the capillaries.	At the arteriole end the hydrostatic pressure is higher than the oncotic pressure, so fluid is forced out through the capillary walls. Large plasma proteins stay in the blood. At the venous end, water returns by osmosis down a water potential gradient, and the excess drains into the lymphatic system.
Cardiac cycle: atrial systole	The heart squeezes.	The atria contract, the pressure rises and the atrioventricular valves open, so blood is forced into the ventricles.
Cardiac cycle: ventricular systole	The valves open and close.	The ventricles contract and the pressure rises above the atria, so the atrioventricular valves close. When the ventricular pressure exceeds the pressure in the aorta, the semilunar valves open and blood is forced out.
Cardiac cycle: diastole	The heart relaxes.	The pressure in the aorta is higher than in the ventricle, so the semilunar valves close. The atria fill with blood and the pressure builds again.
Cardiac output	How much blood the heart pumps.	Cardiac output = stroke volume multiplied by heart rate, in cm3 per minute.
Haemoglobin loading and unloading	Haemoglobin picks up oxygen and drops it off.	At a high partial pressure of oxygen in the lungs haemoglobin loads oxygen to form oxyhaemoglobin. At a low partial pressure in the tissues it unloads oxygen.
S-shaped dissociation curve	It's a curve.	Binding of the first oxygen molecule changes the shape of haemoglobin, which makes it easier for further molecules to bind. This cooperative binding makes the curve S-shaped.

Concept	Own words (loses marks)	Exam language (scores)
Bohr effect	CO2 changes the haemoglobin.	A high concentration of CO2 lowers the pH. Haemoglobin changes shape and has a lower affinity for oxygen, so the curve shifts to the right and more oxygen is unloaded at the respiring tissues.
Fetal haemoglobin	The baby's blood is stronger.	Fetal haemoglobin has a higher affinity for oxygen than adult haemoglobin at the same partial pressure, so the curve is to the left, which lets it load oxygen from the mother's blood in the placenta.
Carbon dioxide transport	CO2 goes in the blood.	Most CO2 is converted to hydrogencarbonate ions in red blood cells by carbonic anhydrase. Some is carried as carbaminohaemoglobin and a small amount dissolves in the plasma.
Water uptake by roots	Roots take in water.	Water enters root hair cells by osmosis down a water potential gradient. It crosses the cortex by the apoplast pathway (through cell walls) and the symplast pathway (through cytoplasm and plasmodesmata). The Casparian strip blocks the apoplast pathway at the endodermis, forcing water into the symplast.
Cohesion-tension	Water is pulled up the plant.	Evaporation from the mesophyll cells lowers the water potential of the leaf and creates tension in the xylem. Water molecules are held together by hydrogen bonds (cohesion), so a continuous column is pulled up. Adhesion to the xylem walls helps.

Concept	Own words (loses marks)	Exam language (scores)
Factors affecting transpiration	Wind makes plants lose more water.	Light causes stomata to open. A higher temperature increases the kinetic energy of the molecules, so evaporation increases. Low humidity steepens the water potential gradient. Wind removes water vapour from around the leaf, so the gradient stays steep. Each of these increases the rate of transpiration.
Potometer	It measures how much water the plant uses.	A potometer measures the rate of water uptake, which is used as an estimate of transpiration. Measure the distance moved by the air bubble in a set time. Not all the water taken up is lost by transpiration.
Mass flow in the phloem	Sugar is moved around the plant.	Sucrose is actively loaded into the sieve tubes, which lowers the water potential, so water enters by osmosis and the hydrostatic pressure increases at the source. Mass flow carries sucrose to the sink, where it is unloaded and used or stored.
Evidence for mass flow	Scientists have tested it.	Ringing a tree removes the phloem, and sugar builds up above the ring. Aphid stylets sample the sap, which is under pressure. Radioactive tracers (14C) can be followed along the phloem.

Key terms to include: diffusion distance, concentration gradient, countercurrent, ventilation, spiracle, tracheole, hydrostatic pressure, oncotic pressure, atrioventricular valve, semilunar valve, affinity, cooperative binding, apoplast, symplast, Casparian strip, cohesion, adhesion, tension, transpiration, sieve tube, source, sink, mass flow.

Topic 10: Ecosystems

This topic is the one most likely to need new learning. The sub-topics below follow the usual 9BI0 content; check them against your class notes.

Concept	Own words (loses marks)	Exam language (scores)
Ecosystem, habitat, population, community	Where animals live.	An ecosystem is all the living organisms (biotic) and the non-living factors (abiotic) in an area. A habitat is where an organism lives, a population is all the members of one species in a habitat, and a community is all the populations in a habitat.
Niche	Where it lives.	The role of a species in its ecosystem, including its habitat, feeding relationships and adaptations. Two species cannot occupy exactly the same niche in the long term.
Abiotic and biotic factors	Things in the environment.	Abiotic factors are non-living, such as temperature, light intensity and pH. Biotic factors are living, such as competition, predation and disease.
Interspecific and intraspecific competition	Animals fight over food.	Interspecific competition is between different species for the same resources. Intraspecific competition is between members of the same species and has a greater effect on population size, because they use exactly the same resources.

Concept	Own words (loses marks)	Exam language (scores)
Predator and prey cycles	Predators eat prey and numbers go up and down.	As the prey population increases, more food is available, so the predator population increases. More predation reduces the prey population, so the predator population then falls because of less food. The cycle repeats, with the predator peak lagging behind the prey peak.
Carrying capacity	The most animals that can live there.	The maximum population size that an environment can support over a long period, limited by resources such as food, space and disease.
Limiting factors	Things that stop the population growing.	Density-dependent factors, such as food supply and disease, have a greater effect as the population gets larger. Density-independent factors, such as fire and extreme weather, affect the population regardless of its size.
Population growth curve	The population goes up then flattens.	Lag phase (slow growth, few individuals), exponential phase (births exceed deaths, resources plentiful), stationary phase (births equal deaths at the carrying capacity), then decline when resources run out.
Primary succession	Plants take over a bare area.	Pioneer species such as lichens colonise bare rock. They tolerate harsh abiotic conditions, and as they die and decompose they form soil. This changes the abiotic conditions so that other species colonise, with an increase in species diversity and biomass.

Concept	Own words (loses marks)	Exam language (scores)
Climax community	The final stage of succession.	A stable community in balance with the abiotic conditions, with the greatest diversity and biomass, and no further change in species composition.
Secondary succession	It happens after a fire.	Succession on land where soil is already present, such as after a fire or the abandonment of farmland. It is faster than primary succession because the soil already contains nutrients and seeds.
Quadrats	Count the plants in a square.	Place quadrats at random using a random number generator for coordinates, to avoid bias. Calculate percentage cover or frequency, and repeat with enough quadrats to get a representative mean.
Transects	Walk in a line.	A line or belt transect is used to sample along an environmental gradient, such as from the shore inland. Place quadrats at regular intervals and record the species present.
Mark-release-recapture	Catch, mark and count animals.	Capture a sample (n1) and mark the animals without harming them, release them, then capture a second sample (n2) and count the marked ones (m2). Population estimate = (n1 multiplied by n2) divided by m2.

Concept	Own words (loses marks)	Exam language (scores)
Assumptions of mark-release-recapture	The marks stay on.	The marked individuals mix randomly with the population, there is no migration, births or deaths in the time between samples, and the mark is not lost or does not make the animal more likely to be caught or eaten.
Gross and net primary productivity	How much energy plants make.	Gross primary productivity (GPP) is the total energy fixed by photosynthesis. Net primary productivity (NPP) = GPP minus the energy lost in respiration (R), and is the energy available to the next trophic level.
Energy transfer between trophic levels	Energy is lost along the food chain.	Only a small percentage is transferred because energy is lost as heat from respiration, in uneaten parts, and in faeces and urine. Efficiency = energy transferred to the next level divided by energy available at the previous level, multiplied by 100.
Carbon cycle	Carbon goes round and round.	Carbon is fixed by photosynthesis, passes along food chains by feeding, and is returned to the atmosphere by respiration, and by decomposition (saprobionts) and combustion.

Concept	Own words (loses marks)	Exam language (scores)
Nitrogen cycle	Bacteria change nitrogen.	Nitrogen fixation converts N2 to ammonium (Rhizobium in root nodules). Ammonification by saprobionts converts nitrogen in dead matter to ammonium. Nitrification converts ammonium to nitrite (Nitrosomonas) then nitrate (Nitrobacter). Denitrification converts nitrate to N2 in waterlogged, anaerobic soil.
Saprobionts	Bacteria and fungi rot dead things.	Saprobionts secrete enzymes onto dead organic matter (extracellular digestion) and absorb the soluble products. This recycles nutrients to the soil.
Eutrophication	Fertiliser pollutes rivers and kills the fish.	Nitrate from fertiliser run-off causes rapid algal growth, and the algal bloom blocks light, so plants die. Aerobic bacteria decomposing the dead plants use up the oxygen, so the biological oxygen demand rises and fish and other aerobic organisms die.
Greenhouse effect	Gases warm the planet.	Greenhouse gases such as CO2 and methane absorb infra-red radiation reflected from the Earth's surface and re-emit it, which warms the atmosphere. Increases in these gases enhance the effect and contribute to global warming.

Concept	Own words (loses marks)	Exam language (scores)
Human impact on ecosystems	Humans destroy habitats.	Deforestation and monoculture reduce habitat variety and species diversity. Overfishing lowers population sizes, and pollution changes the abiotic factors.
Sustainability	Don't use up too much.	Resources are used at a rate that allows them to be replaced, for example fishing quotas, larger mesh sizes, closed seasons and crop rotation.
In situ conservation	Protect the animals where they live.	The species is protected in its natural habitat, for example in a national park. The natural behaviour and habitat are maintained, but poaching and disease are difficult to control.
Ex situ conservation	Keep the animals in a zoo.	The species is protected outside its natural habitat, for example captive breeding or seed banks. Stud books manage genetic diversity, but small populations risk inbreeding, and reintroduction may fail if the habitat has been lost.

Key terms to include: abiotic, biotic, niche, interspecific, intraspecific, carrying capacity, density-dependent, pioneer species, climax community, random sampling, transect, GPP, NPP, trophic level, saprobiont, nitrification, denitrification, eutrophication, biological oxygen demand, sustainable, in situ, ex situ.

Skills: data, calculations, practicals, statistics and long answers

Describing data and graphs

Situation	Own words (loses marks)	Exam language (scores)
Describe a trend	It goes up and then down.	As [x] increases from [a] to [b] [unit], [y] increases from [c] to [d] [unit]. Above [b], [y] decreases to [e] [unit]. Quote the figures at the start, the peak and the end.
Compare two sets of data	A is higher than B.	State the difference with figures: A is 30 cm3 compared with 12 cm3 for B, which is 2.5 times greater. Mention any overlap in the error bars or ranges.
Draw a graph	Plot the points and join them.	The independent variable goes on the x-axis and the dependent variable on the y-axis. Label both axes with the quantity and units, use a linear scale that fills more than half the grid, plot points accurately, and join with a smooth line or a ruled line of best fit as the data requires.
Explain a plateau	It stops going up.	State the limiting factor: for enzymes, all the active sites are saturated; for photosynthesis, another factor such as light or CO2 has become limiting.

Calculations

Quantity	Formula	Exam reminder
Magnification	image size divided by actual size	Convert to the same units first; 1 mm = 1000 µm
Percentage change	(final minus initial) divided by initial, multiplied by 100	State a positive or negative sign
Rate	change in quantity divided by time	Give units such as cm3 per second
Mitotic index	cells in mitosis divided by total cells	Give as a decimal or a percentage as asked

Quantity	Formula	Exam reminder
Index of diversity	D = 1 minus the sum of (n/N) squared	Calculate each (n/N) squared first and show them
Population estimate	(n1 multiplied by n2) divided by m2	n1 is the first sample, n2 the second, m2 the marked recaptures
Cardiac output	stroke volume multiplied by heart rate	Check the units of volume and time
Surface area to volume ratio	surface area divided by volume	Calculate for each shape and compare
Efficiency of energy transfer	energy to the next level divided by energy available, multiplied by 100	Give to 2 significant figures unless told otherwise
NPP	GPP minus respiratory losses (R)	State the units, for example kJ per m2 per year

Practical design and evaluation

Idea	Exam language (scores)
Variables	The independent variable is the one you change, the dependent variable is the one you measure, and the control variables are kept the same and are named with the method of control, such as using a water bath for temperature.
Reliability	Repeat the experiment at least three times and calculate a mean, so random errors are reduced and anomalies can be identified.
Validity	Control all the variables other than the independent variable, so the results are due only to the independent variable.
Accuracy and precision	Accuracy is how close a measurement is to the true value; precision is how close repeated measurements are to each other.

Idea	Exam language (scores)
Improving a method	Use a more precise measuring instrument, such as a gas syringe rather than a measuring cylinder, take readings at more intervals, or increase the sample size.
Safety	Name the specific hazard and the specific precaution, for example wear eye protection when handling hydrogen peroxide.
Control experiment	A control with boiled enzyme (or no enzyme) shows that the change is caused by the enzyme and not by another factor.

Statistical tests

Test	Use it when	Write this
Chi-squared	Comparing observed with expected frequencies of categories	The null hypothesis states there is no significant difference between the observed and expected frequencies, and any difference is due to chance. If the calculated value is greater than the critical value at p = 0.05, reject the null hypothesis.
Spearman's rank	Testing for a correlation between two sets of paired data	The null hypothesis states there is no significant correlation between the two variables. If the calculated value is greater than the critical value at p = 0.05, reject the null hypothesis; there is a significant correlation. Correlation does not prove cause.
Student's t-test	Comparing the means of two sets of normally distributed data	The null hypothesis states there is no significant difference between the means. If the calculated value is greater than the critical value at p = 0.05, reject the null hypothesis; there is a significant difference.

Always read the degrees of freedom or sample size, find the critical value at p = 0.05, compare, then write the conclusion in context.

Six-mark and extended answers

1. Spend one minute listing the six or more separate points before writing. Each is a possible mark.
2. Start with a clear opening sentence that answers the question directly.
3. Write one point per sentence, using key terms and a link such as "which means" or "so".
4. For an evaluation, give an advantage and a disadvantage of each option, then finish with a judgement that is justified.
5. Do not repeat a point in new words; the mark scheme gives credit once.
6. For "suggest" questions, apply the biology you know (diffusion, water potential, enzymes) to the new situation described.

Practice routine and checklist

The three-pass routine (15 minutes a day, one topic block at a time)

Pass	What to do	Time
1. Recall	Read the concept, cover the answer column and write the answer from memory in full sentences	5 minutes
2. Compare	Uncover the exam-language answer, then underline every key term that was missing or vague in the written answer	4 minutes
3. Rewrite	Cover the page again and rewrite the answer using the missing terms; speak it aloud once	6 minutes

Work through the topics in this order: Topic 10 first (newest content), then Topic 4, Topic 1, Topic 2, Topic 3. Use the flashcard deck on a phone for short gaps in the day.

Self-check before handing in any answer

- [ ] Does every sentence name the structure, molecule or process, rather than using "it" or "stuff"?
- [ ] Is there a key term for each mark available?
- [ ] Does each "explain" answer give a reason and a consequence, not just a description?
- [ ] For data, have I quoted figures with units?

- [ ] For calculations, have I shown the formula, the substitution and the units?
- [ ] For an evaluation, have I given both sides and a justified judgement?

Progress tracker

After each practice paper, count the marks lost to "imprecise wording" using the diagnostic grid in the mock paper's mark scheme. The aim is to see that number fall paper by paper.